

Answer Key — Biology Grade 11

Sub-Strand 3.1: Reproduction in Animals • Lesson 1: Frogs, Hens and Cows: Many Ways to One Goal – How Do Animals Reproduce?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
B	B	C	D	B	B	A

USE AFTER: PREDICT PHASE — Predict: viewing the photos of frogspawn in a Kisumu wetland pond and a hen on eggs

Q1. Near Kisumu, a learner looks at frogspawn in a pond and a hen's egg in a nest. Which statement correctly describes what is seen?

- A. Both frog eggs and hen's eggs are hard-shelled and laid on land.
- B. Frog eggs are covered in jelly and released into water, while a hen's egg is hard-shelled and laid on land.**
- C. Frog eggs are hard-shelled and laid on land, while a hen's egg is covered in jelly and released into water.
- D. Both frog eggs and hen's eggs are covered in jelly and released into water.

Correct answer: B. The images show jelly-covered frog eggs released into pond water and a hen's hard-shelled egg laid on land. This is only a description of what is seen, not an explanation.

USE AFTER: PREDICT PHASE — Predict: teacher checklist for misconceptions such as 'all animals lay eggs'

Q2. A learner claims that 'all animals lay eggs'. Which animal from the lesson photos does NOT fit this claim?

- A. A frog in a pond near Kisumu
- B. A cow with her calf**
- C. A tilapia in Lake Victoria
- D. A hen sitting on her eggs

Correct answer: B. A cow gives birth to a live calf after months of development inside her body, so the claim 'all animals lay eggs' does not fit. Frogs, hens and tilapia all produce eggs.

USE AFTER: PREDICT PHASE — Predict: prediction 'Why do frogs need thousands of eggs but a cow produces one calf?'

Q3. Which statement matches what is described about the number of young in the anchoring phenomenon?

- A. A frog gives birth to a few live young, while a cow lays one hard-shelled egg.
- B. A cow lays thousands of eggs, while a frog gives birth to one live calf.
- C. A frog releases thousands of eggs into water, while a cow gives birth to one live calf after months of development inside her.**
- D. A frog and a cow both produce thousands of young at one time.

Correct answer: C. Frogs release thousands of jelly-covered eggs, while a cow has one calf after months of internal development. At this stage learners only record this pattern, and why it happens is still an open question.

USE AFTER: OBSERVE PHASE — Part A: observing external fertilisation at the school pond or wetland edge

Q4. Your group is going to observe frogspawn at the edge of a school pond. Which rule is correct?

- A. Collect the spawn in your pockets so you can study it in class.
- B. Wade into the pond and pick up the spawn to count the eggs.
- C. Touch the spawn with bare hands, but only for a short time.
- D. Observe from the edge, do not touch the spawn with bare hands, and wash your hands afterwards.**

Correct answer: D. The lesson rules are to observe from the edge, not to touch spawn with bare hands, and to wash hands afterwards.

USE AFTER: OBSERVE PHASE — Part A: recording where eggs are released, jelly coating and number of eggs in the observation table

Q5. A group is filling in its Part A observation table for frog spawn. Which entry is an observation, not an explanation?

- A. Water helps the sperm to reach the eggs.
- B. The eggs are released into the water and are surrounded by a jelly layer.**
- C. The jelly stops the eggs from drying out.
- D. Frogs need thousands of eggs because many are eaten.

Correct answer: B. Saying where the eggs are and what surrounds them is a direct observation. The other entries claim reasons for what is seen, and the table is for recording what is seen.

USE AFTER: OBSERVE PHASE — Part B: sorting animals into oviparous, viviparous and ovoviviparous columns on the classification chart

Q6. In Part B, a group sorts a hen and a cow on the classification chart. Which placement is correct?

- A. Hen: ovoviviparous; cow: oviparous
- B. Hen: oviparous (eggs laid, development outside the body); cow: viviparous (young develop inside the mother and are born live)**
- C. Hen: viviparous; cow: ovoviviparous
- D. Hen: viviparous; cow: oviparous

Correct answer: B. A hen lays eggs that develop outside her body, so it is oviparous. A cow carries the young inside her and gives birth to a live calf, so it is viviparous.

USE AFTER: OBSERVE PHASE — Part B: sorting shark/guppy into the ovoviviparous column on the classification chart

Q7. A source says that in a female guppy the eggs develop and hatch inside her body, and she does not feed them through a placenta-type connection. In which column should the group place the guppy?

- A. Ovoviviparous, because eggs develop and hatch inside her body without placenta-type feeding**
- B. Oviparous, because the eggs hatch
- C. Viviparous, because the young develop inside her
- D. Oviparous, because it produces eggs

Correct answer: A. Ovoviviparous animals keep the eggs inside the body where they develop and hatch, without placenta-type nourishment from the mother. This matches the guppy description.